

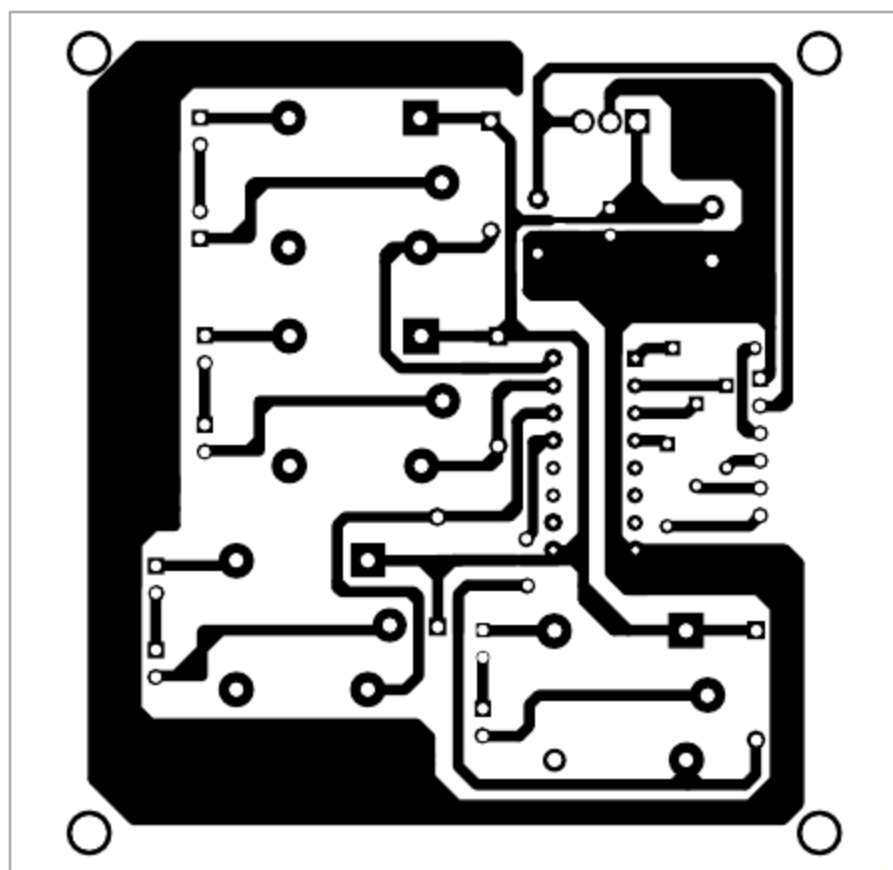


Fig. 4: Actual-size PCB layout of the device control

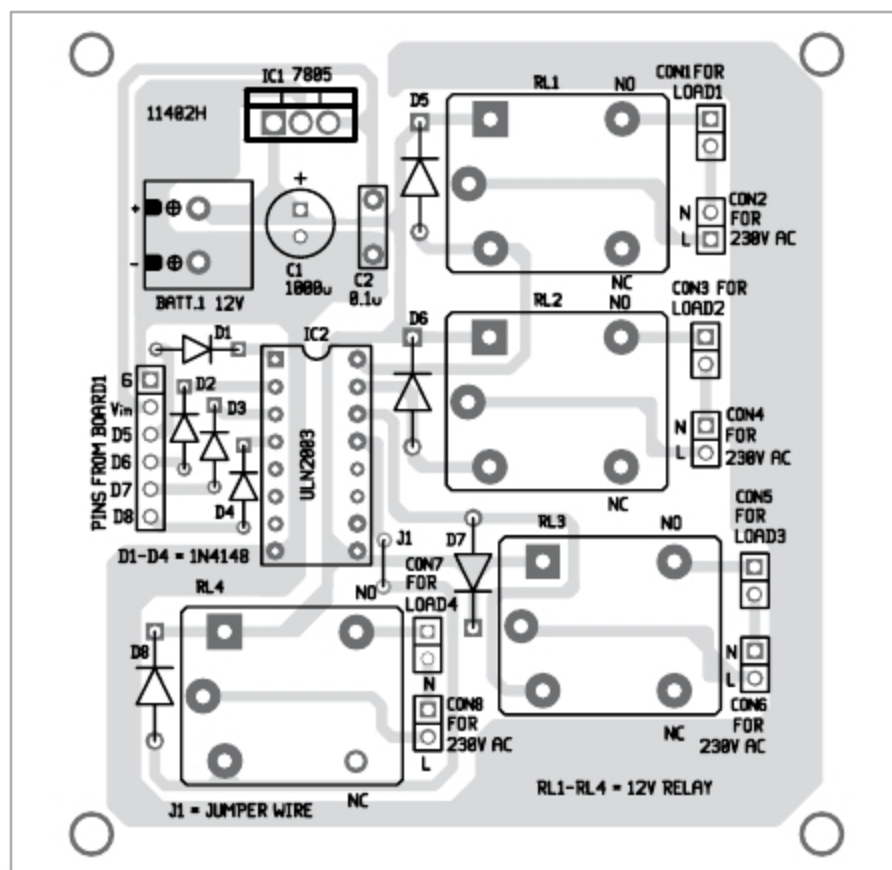


Fig. 5: Components layout for the PCB



Fig. 6: Login page in STA mode

the instructions given in the program code. To login, fill in the user name as admin and password as admin, and click on Login. (You can customise the user name and password in the program code and use these details during login.) Now, you will be able to switch on or switch off the device. Fig. 8 shows the control panel with device status.

In case of invalid login credentials, the program will display an error message along with the number of attempts. If invalid login attempts equal 10, or value of noOfTry defined in the code, the program will lock down for some time as defined by lockDownTime in the code. After

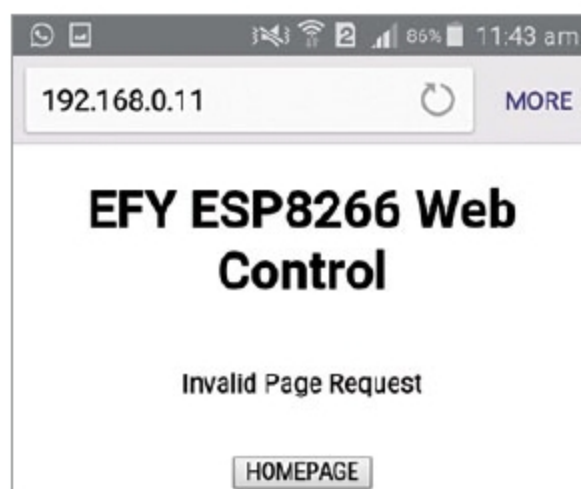


Fig. 7: Invalid page

#### PARTS LIST

##### Semiconductors:

- Board1 - NodeMCU ESP8266 12E
- IC1 - 7805, 5V voltage regulator
- IC2 - ULN2003, relay driver IC
- D1-D4 - 1N4148, signal diode
- D5-D8 - 1N4007, rectifier diode

##### Capacitors:

- C1 - 1000µF, 35V electrolytic
- C2 - 0.1µF ceramic disk

##### Miscellaneous:

- BATT.1 - 12V battery
- CON1-CON8 - 2-pin connector
- 2-pin terminal connector for battery
- 6-pin connector for Board1
- RL1-RL4 - 12V, single- changeover relay



Fig. 8: Control panel showing device status

preset lock-down time, the program will automatically unlock and you can try again with valid credentials. **EFY**

#### EFY Note

The source code of this project is available for free download at [source.efymag.com](http://source.efymag.com)



David Hembram is an electronics hobbyist. His interests include circuit designing, PCB designing, programming embedded systems, Arduino, PIC and Raspberry Pi